

17. Unforced tuned mass damper

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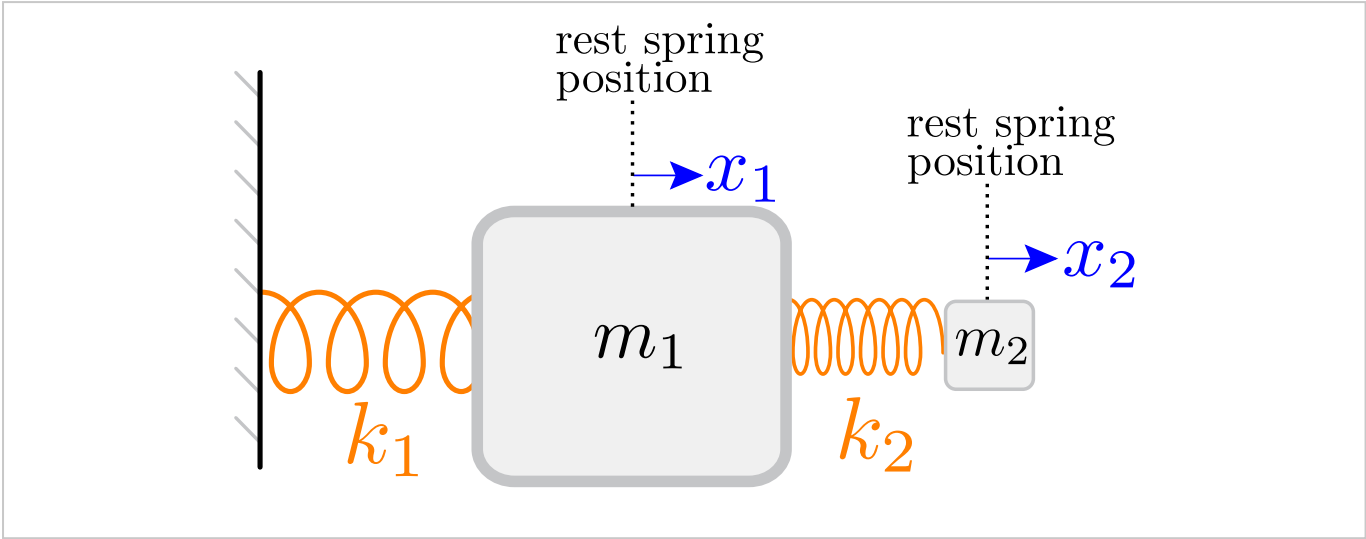
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Angular frequencies for the unforced tuned mass damper

1.0/1 point (graded)

Recall from a previous video a swaying building with a tuned mass damper installed can be modelled as a forced coupled oscillator.



The companion system, in the case when $\mathbf{F} = \mathbf{0}$, and when no dashpots are present ($\mathbf{b}_1 = \mathbf{b}_2 = \mathbf{0}$), can be written as

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{k_1+k_2}{m_1} & \frac{k_2}{m_1} & 0 & 0 \\ \frac{k_2}{m_2} & -\frac{k_2}{m_2} & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}. \tag{4.105}$$

This can also be obtained from the previous coupled oscillator by setting $k_3 = 0$.

Set the parameters to $m_1 = 100$, $m_2 = 1$ and $k_1 = 100$, $k_2 = 1$, to model the facts that the building is much heavier than the mass in the tuned mass damper, and the building is much stiffer as a spring than the one in the tuned mass damper. We will now follow the procedure as in the text above to find the pure sinusoidal modes of this system.

First, rewrite the above system in a more compact form:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \quad \text{where} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ and } \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}. \tag{4.106}$$

Write down the matrix $\mathbf{B}$.

(Enter `[[a,b],[c,d]]` for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

$\mathbf{B} =$ ✔ Answer: `[[ -101/100, 1/100 ], [ 1, -1 ]]`

Solution:

With $m_1 = 100$, $m_2 = 1$ and $k_1 = 100$, $k_2 = 1$, the companion system becomes

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{101}{100} & \frac{1}{100} & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}, \tag{4.107}$$

which can be rewritten more compactly as

$$\frac{d}{dt} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \quad \text{where} \tag{4.108}$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \text{ and} \tag{4.109}$$

$$\mathbf{B} = \begin{pmatrix} -\frac{101}{100} & \frac{1}{100} \end{pmatrix}. \tag{4.110}$$

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Find the eigenvalues and angular frequencies

4/4 points (graded)

Continuing from the problem above, find the eigenvalues of **B**.

$\lambda_1 =$ ✓ Answer: -201/200+sqrt(401)/200

$\lambda_2 =$ ✓ Answer: -201/200-sqrt(401)/200

What are the angular frequencies ω_1, ω_2 of the two pure sinusoidal modes of unforced tuned mass damper?

$\omega_1 =$ ✓ Answer: (sqrt(401)-1)/20

$\omega_2 =$ ✓ Answer: (sqrt(401)+1)/20

Solution:

The eigenvalues of **B** are the roots of the equation:

$$\left(\lambda + \frac{101}{100}\right)(\lambda + 1) - \frac{1}{100} = \lambda^2 + \left(2 + \frac{1}{100}\right)\lambda + 1 = 0 \quad \Rightarrow \quad (4.111)$$
$$\lambda = -\frac{201}{200} \pm \frac{\sqrt{401}}{200}.$$

Note that both of these eigenvalues are negative.

The eigenvalues of $\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix}$ are the square roots of the eigenvalues of **B** and so are purely imaginary:

$$\pm i \frac{1}{10} \sqrt{\frac{201 - \sqrt{401}}{2}} = \pm i \frac{1}{10} \sqrt{\frac{401 + 1 - 2\sqrt{401}}{4}} \quad (4.112)$$

$$= \pm i \frac{1}{10} \sqrt{\frac{(\sqrt{401} - 1)^2}{4}} \quad (4.113)$$

$$= \pm i \frac{\sqrt{401} - 1}{20} \quad (4.114)$$

$$\pm i \frac{1}{10} \sqrt{\frac{201 + \sqrt{401}}{2}} = \pm i \frac{\sqrt{401} + 1}{20} \quad (4.115)$$

Therefore, the angular frequencies of the pure sinusoidal modes are $\frac{\sqrt{401} - 1}{20}$ and $\frac{\sqrt{401} + 1}{20}$.

Note: In the homework, you will find the normal modes of coupled oscillators with dashpots.

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